



Long-lived volcanism expressed through mare infilling, domes and IMPs in the Arago region of the Moon

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ABSTRACT

Mare Tranquillitatis corresponds to the deposit of successive Early to Late Imbrian basaltic units filling the Tranquillitatis basin on the Moon. The present study focuses on the western half of the mare, in the vicinity of the Arago crater (6.16°N, 21.42°E). High resolution datasets from recent remote sensing missions were used to reconstruct the geologic history of the area, which includes a variety of geological features such as: 8 extrusive domes, numerous wrinkle ridges, a sinuous rille and 37 Irregular Mare Patches (IMPs). We performed crater counting to date the domes emplacement and estimated the domes lava rheologic properties (plastic viscosities, lava effusion rates, and durations of effusion) using their geometric characteristics. As a result we classify the Arago domes into three groups: E1-type domes (Arago 1 and 8), H1-type domes (Arago 4 to 7), and B-type domes (Arago 2 and 3) respectively emplaced ~ 3.7 Ga, ~ 3.4 Ga, and ~ 2.8 Ga ago. IMPs are observed in the younger mare unit and on the top of the Arago 6 dome; they likely correspond to a late stage of waning mare volcanism in the area. Both IMPs and domes have a composition similar to the surrounding mare in the VNIR spectral domain, consistent with mafic materials. The exceptionally long-lived volcanism and its diversity recorded in the Arago region may be related to both a Th-rich anomaly reported nearby and to the large-scale magma center responsible for the Lamont positive Bouguer anomaly. In addition, volcanic features of the Arago region are superimposed on a Ti-rich mare unit visited by the Apollo 11 crew 175 km to the south at Tranquility Base. The geological and historical richness of this region makes it a compelling site for future science and/or In Situ Resources Utilization (ISRU) driven missions to the Moon.

1. Introduction

Recent studies show that the geologic and thermal evolution of the Moon is far more complex than previously thought (e.g., Wilson and Head, 2003, 2017; Braden et al., 2014; Zhao et al., 2017). This complexity is well illustrated by the diversity of volcanic landforms that are observed in the vicinity of the Arago crater, located in the western part of Mare Tranquillitatis. Mare Tranquillitatis is composed of Early to Late Imbrian geological units filling most of the ~ 800 km diameter impact basin of the same name with extensive basaltic flows, up to ~ 3.4 Ga (Pieters, 1978, Head and Wilson, 1992; Staid et al., 1996, 2011; Hiesinger et al., 2000, 2011). The study area encompasses the western part of Mare Tranquillitatis (18–30°E, 2 to 12 °N; Fig. 1a), in the vicinity of Arago and Carrel craters, 175 km north of Tranquility Base, the Apollo

11 landing site (Fig. 1c). Similarly to the Apollo 11 landing site area, the Arago region is dominated by relatively dark, ancient (>3.6 Ga), high-Ti (>9 wt% TiO₂) mare basalt deposits (e.g., Pieters, 1978). The study area covers four of the geologic units previously mapped by Kodama and Yamaguchi (2003) and labeled Tr1 (dated at 3.85 Ga) through Tr4 (dated at 3.59 Ga), the later being the same than the Tranquility Base unit. Estimated FeO and TiO₂ contents increase from Tr1 to Tr4 (Kodama and Yamaguchi, 2003).

Eight volcanic domes and 37 Irregular Mare Patches (IMPs) have been reported in this area (Lena et al., 2013; Braden et al., 2014). The Arago domes were previously studied by Lena et al. (2013) and catalogued into two classes, based on their morphometric and spectral characteristics. Both classes of domes have low slope values and roughly circular outlines, and were therefore interpreted as likely emplaced by extrusive, effusive

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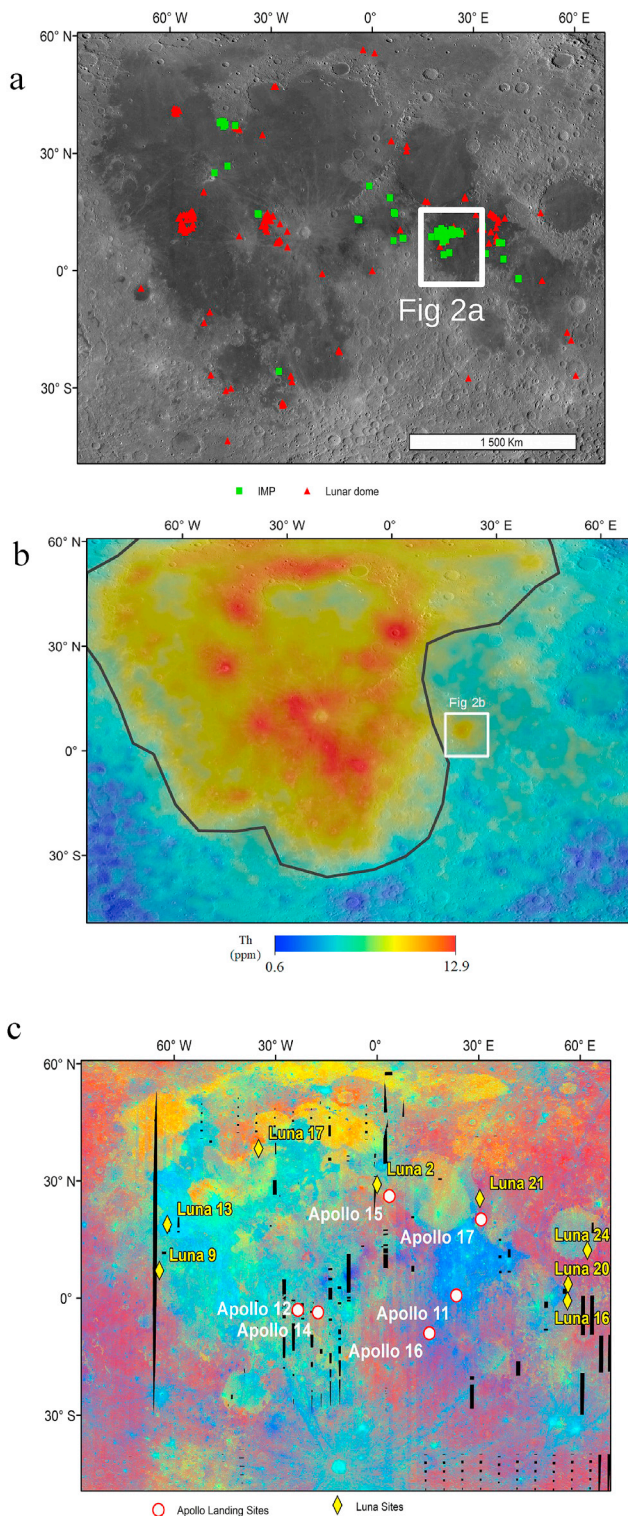


Fig. 1. A. Location of the study area (white box) on the nearside of the Moon. Previously referenced lunar domes and IMPs are shown as red triangles and green squares respectively. The LROC WAC global mosaic is used as a background. B. Location of the study area (white box) on the Lunar Prospector Thorium abundance map. The outline of the Procellarum KREEP Terrane (PKT) is represented as a thick black line. C. Location of the study area (white box) on Clementine RGB mineral composite (R = 750/415 nm, G = 750/950 nm, B = 415/750 nm, see text for more details). Apollo and Luna landing sites are labeled.

volcanism (Lena et al., 2013). IMPs were recently identified based on their high albedo and paucity of craters, and are tentatively interpreted as recent volcanic units (<1 Ga, Braden et al., 2014) and/or the result of ancient eruptions of magmatic foams (>3.8 Ga, e.g., Wilson and Head, 2017; Qiao et al., 2018). Additional volcanic and tectonic features in the study area include a 200 km long sinuous rille (named Diamondback Rille), numerous north-south-trending wrinkle ridges, and extensional grabens bordering the highlands, indicating a complex geologic history. Adding to the complexity, the region shows: i) a positive gravity anomaly centered 60 km southeast of Arago crater (tentatively interpreted as a buried impact structure; and referred to as the “Lamont” anomaly (Evans et al., 2016; Zhang et al., 2018) and ii) a high-Th content up to 6.1 ppm, which represents an anomaly for a region located outside the Procellarum KREEP Terrane (PKT) (e.g., Jolliff et al., 2000; Lawrence et al., 2003, Fig. 1b).

The present study describes the characteristics and relative timing of the emplaced volcanic features using a variety of new high resolution datasets. We show that the Arago region is not only a promising site for science because of its long and complex geologic history, but also a potential region of interest for future In Situ Resources Utilization (ISRU) missions.

2. Datasets and methods

2.1. Remote sensing datasets

A number of datasets from the Planetary Data System and SELENE archive available between 18 and 30°E, and between 3 and 12°N were processed and integrated into a Geographic Information System (GIS). The data collection includes: Kaguya Terrain Camera (TC) panchromatic images and associated Digital Terrain Model (DTM, spatial resolution of 10 m/px), both assembled in mosaics (Haruyama et al., 2008); Lunar Reconnaissance Orbiter Camera Wide Angle Camera (LROC WAC, resolution of 100 m/px) and Narrow Angle Camera imagery (NAC, resolution < 1 m/px) (Robinson et al., 2010); Lunar Orbiter Laser Altimeter (LOLA) global Digital Elevation Model (DEM, resolution of 118 m/px) (Smith et al., 2017); Gravity Recovery And Interior Laboratory (GRAIL) gravity maps (Zuber et al., 2013); Lunar Prospector Gamma Ray Spectrometry maps (GRS, resolution from 15 to 150 km/px) (Lawrence et al., 1998); Diviner Christiansen Feature position map (Greenhagen et al., 2010) and Clementine UVVIS color ratio mineral map as well as Clementine-derived FeO and TiO₂ maps (McEwen et al., 1994; Lucey et al., 2000a, b; Heather and Duncan, 2002). Kaguya Multiband Imager data (MI, resolution from 10 to 62 m/px) were also used to build a false-color map mimicking the Clementine color ratio mineral map using following bands: R = 750/415 nm, G = 750/950 nm, B = 415/750 nm (e.g., Heather and Duncan, 2002). Moreover, the Moon Mineralogy Mapper (M³) VNIR hyperspectral data were processed with the method of Martinot et al. (2018) to provide mineralogical information for the study area. The M³ spectroscopic data ranges from 430 to 3000 nm with 85 spectral channels and a spatial resolution from ~140 to 280 m/px (Pieters et al., 2009). This study makes use of geometrically, photometrically, radiometrically and thermally corrected M³ Level 2 data (Boardman et al., 2011; Clark et al., 2011; Green et al., 2011; Besse et al., 2013) from the OP1A and OP2C optical periods, which cover dome A7 and domes A3, A5 and A6 respectively. Because space weathering is known to weaken mineralogical absorption bands and to lower reflectance values at shorter wavelengths (inducing a red slope in the VNIR spectrum) (e.g., Adams and McCord, 1970; Pieters et al., 1993, 2000), the spectral continuum was removed following the method of Horgan et al. (2014) and Martinot et al. (2018). Both M³ original and continuum-removed spectra were visually analyzed for definitive mineral identification, performed by comparing M³ spectra with the Reflectance Experiment Laboratory (RELAB) reference library (<http://www.planetary.brown.edu/relab/>). Spectral parameters such as band centers, band depths, band areas, and band

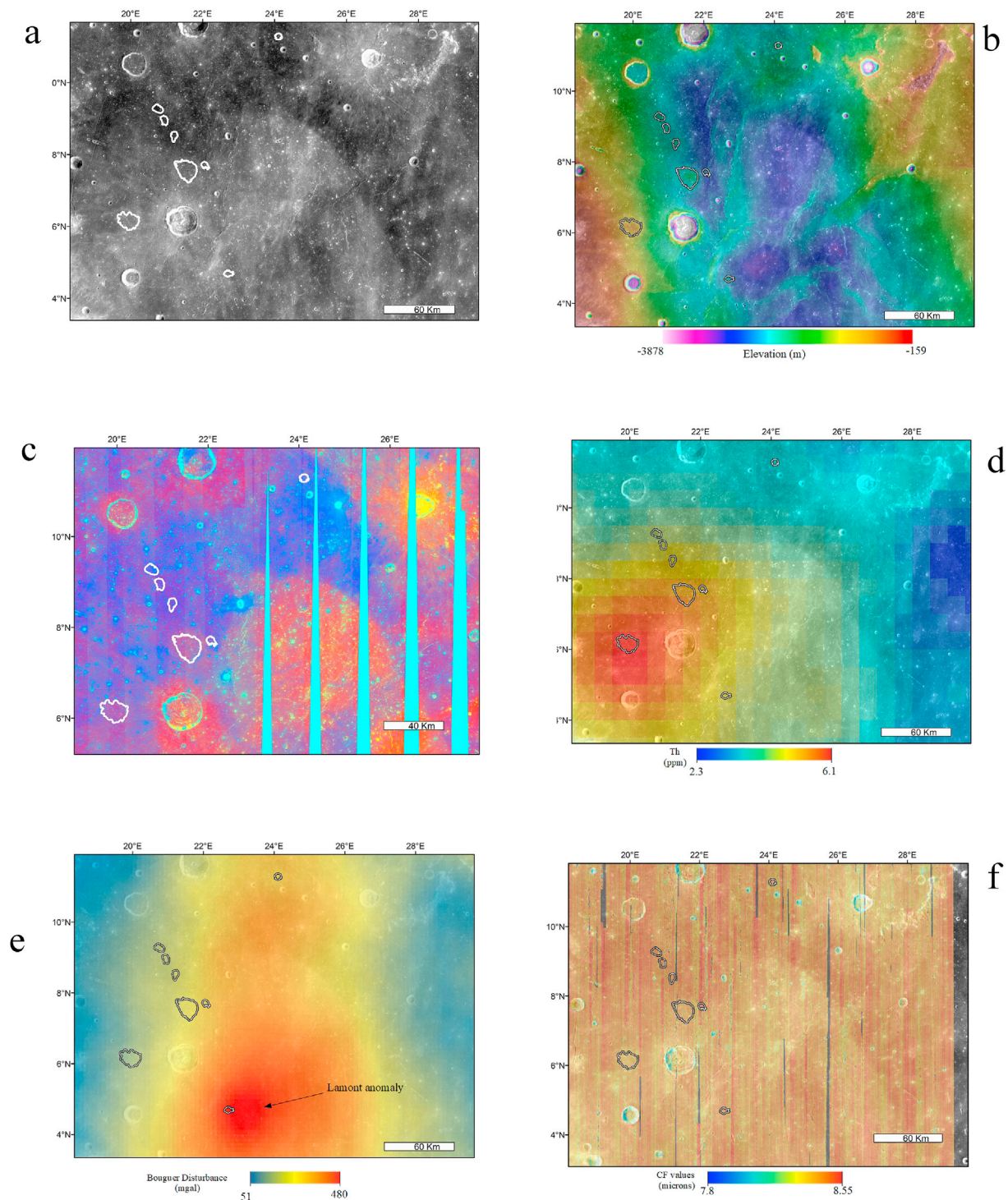


Fig. 2. A. Mosaic of Kaguya Terrain Camera (TC) panchromatic images of the study area. The studied domes are represented as white outlines. B. Digital Terrain Model derived from the Kaguya TC map of the area superimposed on the corresponding TC images. C. False-colour mosaic of Kaguya Multiband Imager using the following band ratios $R = 750/415$ nm; $G = 750/950$ nm; $B = 415/750$ nm. This color composite uses similar bands as the Clementine color ratio mineral map, but offers a higher spatial resolution. D. Lunar Prospector Thorium abundance map of the study area. E. GAIL Bouguer anomaly of the study area. The location of the Lamont anomaly is indicated with an arrow. F. Diviner Christiansen Feature map of the study area.

asymmetries were calculated for the 1 and 2 μ m absorption bands for each spectrum (Martinot et al., 2018). Colour composite maps generated using these parameters allow for visual detection of minerals and high-light potential spatial heterogeneities within the study area.

Previous detections of Irregular Mare Patches from Braden et al. (2014)

and of the lunar domes from Wöhler et al. (2007) were added to the data collection. Both the USGS geological map L-1162 (Fortezzo et al., 2013, renovation of the Wilhelms (1979) map) and the map of Kodama and Yamaguchi (2003) were used to map mare unit boundaries. All data are presented in equidistant cylindrical projection, with the North to the top.

2.2. Modelling the dome rheological properties and ages

Kaguya TC images (Fig. 2a) and their associated DTMs (Fig. 2b) were used to determine the dimensions and geometry of the domes. Height and mean radius were measured to calculate slopes, areas, and volume in order to infer their rheologic properties. As domes do not always lie on a flat surface, but often straddle across topographic breaks, a minimum and a maximum height were measured and their average value was used for rheologic calculations. Crater count was performed on the domes to obtain age estimates by fitting the distribution of the largest craters, using *craterstats 2* software in ArcGIS (Michael and Neukum, 2010). Dome dimensions were entered into rheologic equations to assess the physical properties of the feeding lava flows (e.g., Wilson and Head, 2003; Wohler et al., 2007; Lena et al., 2007, 2013). First order mathematical relationships (Eqs. (1)–(4)) can be applied to link the geometry of volcanic landforms to the physical properties of their forming magmas (e.g., Hulme, 1974; Lyman et al., 2004). Dome eruption can be considered as the extrusion of a Bingham fluid (the cooling magma) characterized by a yield strength τ (Eq. (1)) and a plastic viscosity η (Eq. (2)). Assuming that the magma spreads out onto a relatively flat surface in all directions from the vent, the dome effusion rate E (Eq. (3)) and the duration of the lava effusion T_e (Eq. (4)) can be estimated.

The yield strength τ (expressed in Pa) is given by:

$$\tau = \frac{0.323h^2\rho g}{D/2} \quad (1)$$

Where ρ is the magma density with a mean value set at 2000 kg m⁻³ as in previous studies (Wilson and Head, 2003; Wöhler et al., 2007; Lena et al., 2013), g is the acceleration due to gravity on the surface of the Moon with a value of 1.63 m s⁻², D is the diameter of the dome in meters) and h is the average height of the dome in meters.

The plastic viscosity η (expressed in Pa.s) can then be estimated using:

$$\eta(\tau) = 6 \times 10^{-4} \tau^{2.4} \quad (2)$$

Although the density of terrestrial basaltic lavas is closer to 2800 kg m⁻³ (e.g., Spera, 2000), previous authors have chosen a lower value as a model input for lunar lavas assuming a vesicular foam magma. Taking a density of 2800 kg m⁻³ only increase the viscosity by a factor of ~2.2, and hence is not affecting the interpretation of the results discussed hereafter.

Assuming that the domes have a parabolic cross-section as the advance of the lava front cools (Wilson and Head, 2003), the lava eruption rate E (expressed in m³.s⁻¹) can be estimated by:

$$E = \frac{0.323^{1/2}300\kappa(D/2)^2}{0.65^{5/2}0.72h} \quad (3)$$

Where κ is the thermal diffusivity of the lava with a value of 10⁻⁶m²s⁻¹ (Wilson and Head, 2003).

The duration of the lava effusion T_e (expressed in seconds) can then be estimated with:

$$T_e = V/E \quad (4)$$

Where V is the dome average volume in km³.

The domes of the Arago region were then classified as a function of their determined morphometric, spectral and rheologic properties.

3. Results

3.1. Mapping and distribution of the domes

Eight candidate volcanic domes, labeled Arago 1 to 8 (A1 to A8) were identified using the DTM derived from the TC images (Fig. 2a, 2b and 3). Among them, Arago 8 (4,69°N; 22,72°E) was not reported before.

All the studied domes are located within the two youngest mare units

mapped by Kodama and Yamaguchi (2003) (Fig. 3). Domes Arago 2 and 3 are located in the Tr3 unit (previously described as medium-Ti unit, dated at 3.67–3.71 Ga (Kodama and Yamaguchi, 2003)), whereas domes Arago 1, 4, 5, 6, 7, and 8 are located in the Tr4 unit (the youngest regional unit having the higher-Ti content, dated at 3.59 Ga). With the exception of dome A1 (Fig. 4a) and A7 (Fig. 4e), the domes are located on a topographic step (Fig. 4b, c, d, and f), where the local elevation decreases from west to east. Dome A1 is located in the vicinity of a 100 km long north-south-trending wrinkle ridge (Figs. 4a and 5a). Besides this example, there is no obvious spatial relationship between domes, wrinkle ridges and sinuous rille (Fig. 3).

Although the SE-NW alignment of domes Arago 1, 2, 4, 5, 6, and 8 does not directly crosscut any regional grabens, it suggests a regional tectonic control (see section 4).

3.2. Morphological characteristics

The domes radius, slope, area and volume that we have determined are reported in Table 1. The Arago domes range from 4.8 to 24 km in diameter, 40–400 m in height and 0.8–3° in slope (Table 1, Fig. 4). Four IMPs, characterized by hummocky and blocky floor terrains (Braden et al., 2014) are observed on the top of dome Arago 6 (Figs. 4d and 5b). Pseudo-circular, shallow depressions of lower depth-to-diameter ratios than the standard values expected for fresh impact craters (between 0.11 and 0.21; e.g., Pike (1974); Stopar et al. (2017)) are present on the summits of Arago 1, 7 and 8 and may represent vents, although evidence for extrusive volcanism is not obvious (Fig. 4a, 4e, and 5a). In addition, there is no obvious evidence supporting that these domes could be intrusive structures. Wöhler et al. (2007) previously classified Arago 1 to 7 as low slope, effusive domes.

3.3. Ages

Crater counting was performed on the domes for estimating their ages (Figs. 6 and 7), spanning over ~ 1 Ga, from ~3.75 to ~ 2.79 Ga. Three groups of ages were derived. The older domes are Arago 1 and 8 (3.75 ± 0.05/0.08 Ga and 3.73 ± 0.10/0.34 Ga, respectively), followed by Arago 4 to 7 (3.19 ± 0.19/0.62 Ga, 3.34 ± 0.11/0.32, 3.68 ± 0.08/0.18 Ga and 3.50 ± 0.08/0.19 Ga, respectively). The younger domes are Arago 2 and 3 (3.03 ± 0.17/0.29 Ga and 2.79 ± 0.28/0.36 Ga, respectively). These age estimates rely on the assumption that the domes are monogenetic, i.e. they were formed during a single volcanic event.

3.4. Rheology

The domes minimum or maximum height, diameter and volume were used as inputs for estimating the dome rheologic properties (see section 2.2). Results are shown in Table 2.

Plastic viscosities η range between 1.57 × 10⁴ and 4.05 × 10⁶ Pa s, and lava effusion rate E estimates range from 43 to 230 m³ s⁻¹, yielding a corresponding duration of lava effusion T_e between 0.27 and 6.36 years.

3.5. Spectroscopy

The spectral data from Clementine (Fig. 1c), MI (Fig. 2c), GRS (Figs. 1b and 2d), Diviner (Fig. 2f) and M³ (Fig. 8) do not reveal significant differences between the composition of the domes and the surrounding mare units. Dark blue tones on both the Clementine and Kaguya RGB composites are consistent with high Ti mature mare materials (Figs. 1c and 2c). Using the Kaguya MI RGB composite (Fig. 2c), we extracted a mean value of each ratio of bands, and compared the values with those of Clementine RGB used by Wöhler et al. (2007) in order to rank these domes into different categories (see section 4.1).

M³ VNIR spectra of the different units are presented in Fig. 8. The most common spectrum observed throughout the area is dominated by wide absorptions bands centered at 1.01 and 2.18 μm, consistent with the

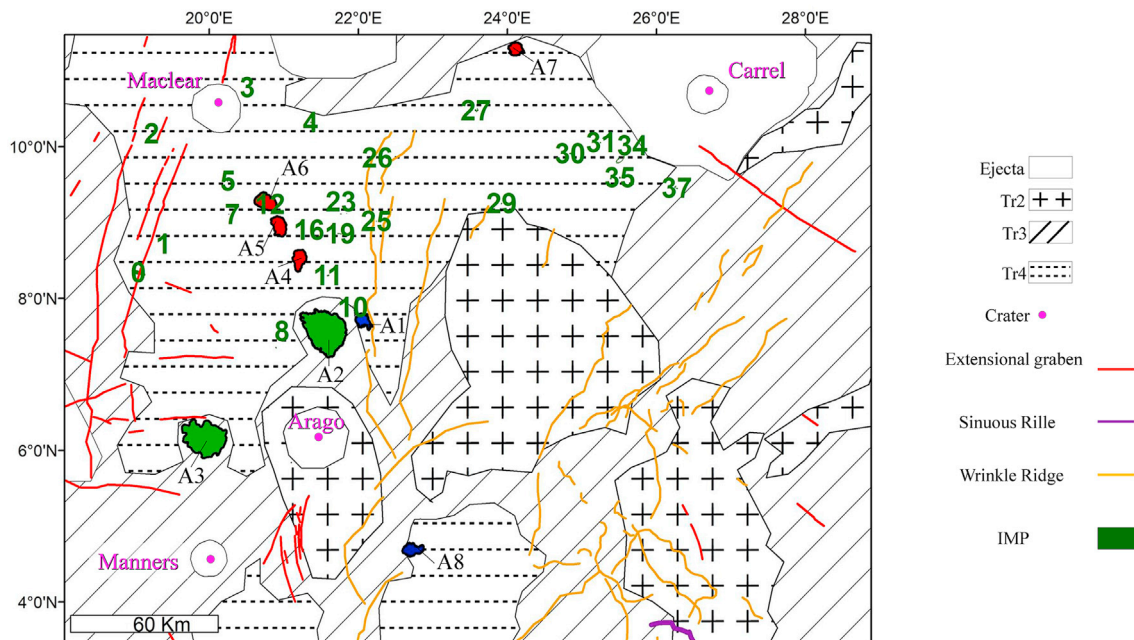


Fig. 3. Interpretative sketch map of the study area: red lines represent extensional grabens; orange lines represent wrinkle ridges; green numbers represent IMPs; units Tr2 to Tr4 from Kodama and Yamaguchi (2003) are also indicated. Domes are represented in different colors according to the classes defined in this paper: green for Arago 2 and 3 of category B; blue for Arago 1 and 8 of category E1; red for Arago 4 to 7 of category H1. The sinuous rille (Diamondback Rille) is represented by a purple line, visible in the southeast of the study area.

presence of high calcium pyroxenes (HCP) such as pigeonite or augite (e.g., Adams, 1974; Cloutis and Gaffey, 1991, Fig. 8c, 8d). Other common lunar minerals such as plagioclase, olivine or spinel (e.g., Martinot et al., 2018) could not be detected throughout the study area. M^3 data do not reveal major compositional differences between the domes covered by the dataset (Arago 3, 5, 6, and 7), neither between the IMPs (#1 and #7) or the surrounding mare units, all of them being also dominated by HCP signatures (Fig. 8c). Spectral parameters maps (Fig. 8b) show that there is little variation in the position of the pyroxene $2\ \mu\text{m}$ absorption center, suggesting that the composition of the area is relatively homogeneous at the M^3 scale (140–280 m/px, depending on the optical period). Small craters exhibit stronger absorption features compared to the surrounding mare or dome material, as they excavate “fresher” material (meaning they were less exposed to space weathering at the lunar surface e.g., a fresh crater in Tr4 in Fig. 8c (blue spectrum) (e.g., McCord and Adams, 1973; Pieters et al., 2000)). Spectral analyses of the largest IMPs (e.g., Sosigenes, IMP#1, sea green spectrum in Fig. 8c) show that they are also characterized by higher band depth values, which indicate that they are optically less mature than the mare and dome materials.

3.6. Dome classification

Based on their geometry (slope value) and their spectral properties (R415/R750), the domes were ranked following the classification scheme by Lena et al. (2013):

- H1: small domes with low slopes ($<2^\circ$) and R415/R750 values between 0.62 and 0.68 (Arago 4 to 7),
- B: large scale domes with higher slopes ($1.3\text{--}5.4^\circ$) and R415/R750 between 0.55 and 0.64 (Arago 2 and 3),
- E1: small domes with higher slopes ($2\text{--}4^\circ$) and R415/R750 between 0.58 and 0.62 (Arago 1 and 8).

4. Discussion

4.1. Revisited dome classification

The use of higher resolution data from Kaguya and LRO in this study permitted to refine the domes' rank into three classes as defined by Lena et al. (2013):

- A2 and A3 fall in the dome category B (instead of class D: low flank slopes, high edifice volume, possibly polygenetic in origin, with higher R415/R750 ratio);
- A4 to A7 fall as either H1 or C2 domes (instead of class A which have a higher R415/R750 ratio). Their dimensions are smaller than C2 domes, and thus match better those of the H1 category. However, it is to be noted that A4 to A7 are not considered polygenetic domes (as previously suggested for some domes of Marius Hills which defined the H class of Lena et al., 2013).
- A1 and the new Arago 8 are in class E1 (instead of class A: lowest slopes but higher R415/R750 ratio).

All the studied domes show relatively low slopes compared to lunar silicic domes reported elsewhere on the Moon (e.g., Wilson and Head, 2003; Wöhler et al., 2006; Glotch et al., 2010, 2011; Lena et al., 2013). In comparison, the silicic domes of Gruithuisen and Mairan (e.g., Wilson and Head, 2003; Glotch et al., 2011) show slopes between 5 and 15° for a similar size range such as Arago 2 and 3 (diameters around 20 km). The low slopes of the Arago domes are more similar to those of other mare domes, such as the shallow domes present on Mons Rümker (Zhao et al., 2017), or the Hortensius, Marius Hills and Milichius domes (Wöhler et al., 2006). The resolution of Diviner data is too low for a good determination of a possible silica composition. However, there is no evidence for low Christiansen Features values/high silica composition on the

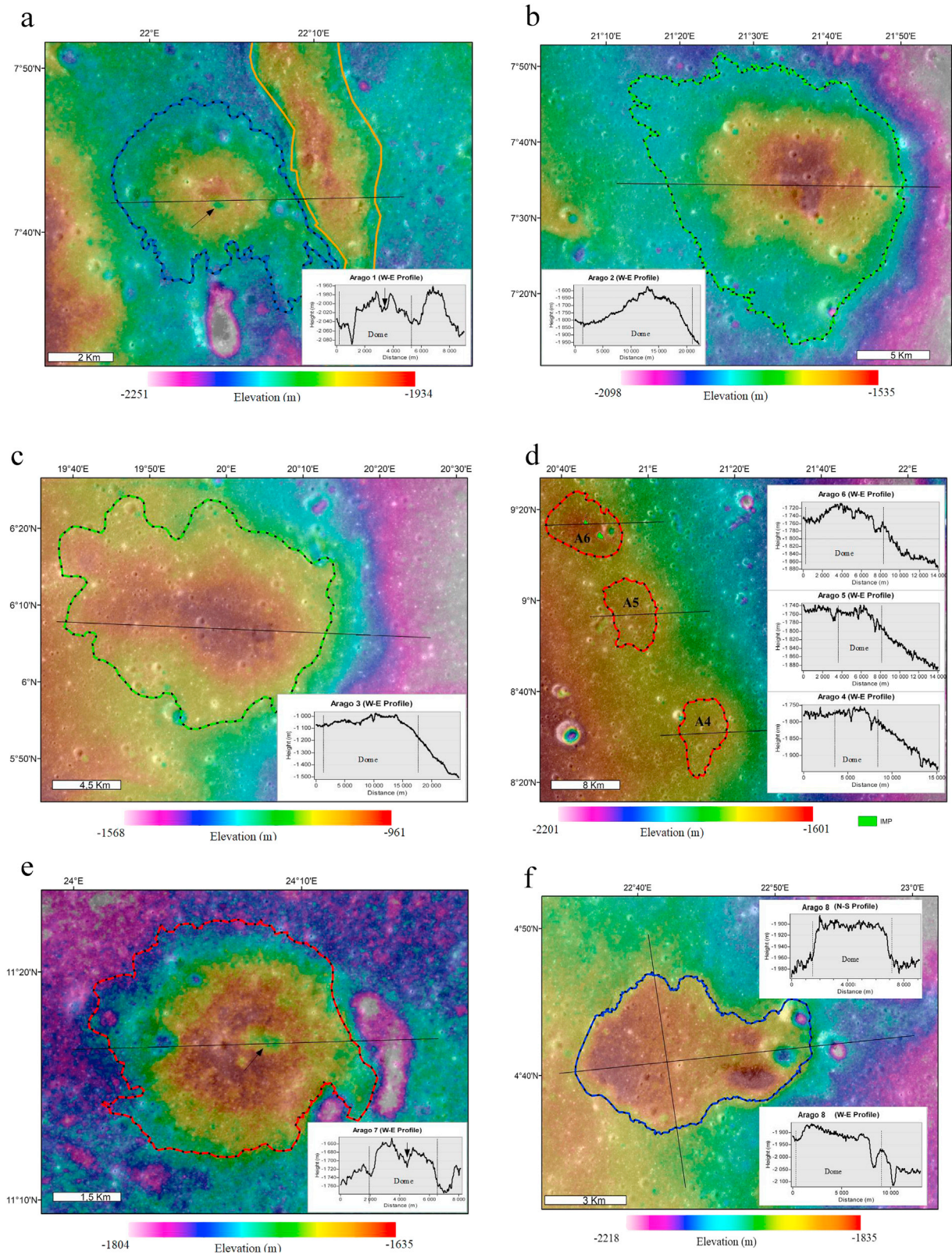


Fig. 4. A to F: Topographic profiles of the Arago 1 to 8 domes. The basemaps are Kaguya TC DTMs (rainbow scale) overlain in transparency on corresponding TC images. Black lines indicate the location of the W-E profiles, except for the new candidate dome Arago 8 where a N-S profile has been added. Local topography seems to control the final shape of the domes.

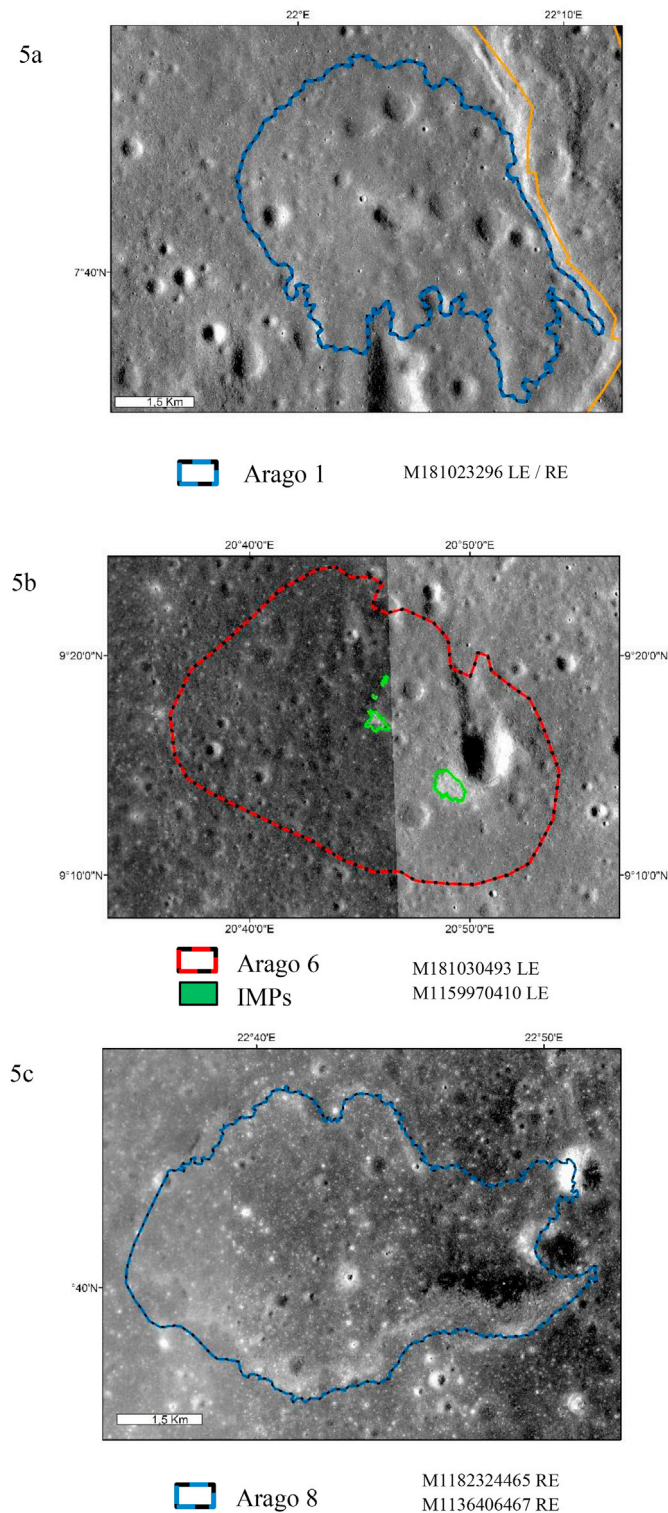


Fig. 5. A. LRO NAC images M181023296LE and RE of Arago 1 (outlined in blue). A wrinkle ridge (brown) appears to reach the base of the dome. B. LRO NAC images M181030493LE and M1159970410LE shows 4 IMPs (outlined in green) on the top of Arago 6 (outlined in red), implying that the IMPs formed after the dome's emplacement. C. LRO NAC images M1182324465RE and M1136406467RE reveal the presence of a previously unmapped dome (outlined in blue), named Arago 8 in this study.

global map from [Greenhagen et al. \(2010\)](#) (Fig. 2f). Both UV and VNIR spectral data indicate that the composition of the domes is similar to the nearby mare unit, i.e. that they are basaltic in composition. Some domes present a rimless crater pit on their summit (interpreted as a volcanic vent); and most domes are roughly circular in shape. Together, these observations confirm the previous assumptions that the Arago domes are likely effusive in nature (e.g., [Lena et al., 2013](#)). The Arago domes might represent a volcanic stage later than the normal (or typical) mare Tranquillitatis volcanism. This is supported by their emplacement on the youngest regional units (Tr4), except for Arago 2 and 3 that are within Tr3 (Fig. 3). Small volumes of remaining magmas could have been emitted at the surface, possibly along fractures, after the major mare flooding episodes. As there is no evidence of high silica contents for any of the domes, the more viscous character of the dome lavas compared to the surrounding mare materials could alternatively be explained by:

- A low magma temperature,
- The presence of more crystals due to progressive cooling in the magma chamber,
- Magmatic differentiation (more evolved compositions),
- Low rate and low volume eruptions,

Or a combination of the above (e.g., [Francis and Oppenheimer, 2004](#), and references therein).

4.2. Dome chronology

Based on previous observations and crater counting-derived chronology, the following chronology for the emplacement of the Arago domes is proposed.

Arago 1 and 8 are respectively dated at $3.75 \pm 0.05/0.08$ Ga and $3.73 \pm 0.10/0.34$ Ga. They would therefore be the first domes to be emplaced. A1 and A8 are emplaced over the Tr4 unit, which was dated at 3.59 Ga by [Jerde et al. \(1994\)](#), and should therefore be younger than Tr4. We consider this discrepancy between estimated ages to be either due to errors in crater counts (the areas counted on the domes are restricted in size and may induce large errors) or in unit mapping. Alternatively, the domes could be more ancient than Tr4 and embayed by the latest regional mare unit. The two domes have, in consideration with all domes of the area, the smaller radii, intermediate heights and volumes, the steepest slopes and are therefore both classified in the E1 category.

Arago 4, 5, 6 and 7 correspond to a later stage of dome activity at ~ 3.68 to 3.19 Ga and are categorized as H1 class domes, characterized by low slopes and small volumes. Their apparent viscosities ($< 2 \times 10^5$ Pa s) yield values lower than Arago 1 and 8 ($> 8 \times 10^5$ Pa s) by an order of magnitude, indicative of less viscous magmas. Domes Arago 4, 5 and 6 are aligned along a SE-NW trend, possibly suggesting post- or syn-formation along tectonic features (e.g., fissure or fault).

Arago 2 and 3 were respectively emplaced at $3.03 \pm 0.17/0.29$ Ga and $2.79 \pm 0.28/0.43$ Ga and represent the latest dome emplacement stage in the region. Arago 2 and 3 have the largest dimensions and show intermediate slopes. Their higher apparent viscosities (4.05×10^6 and 2.75×10^6 Pa s, respectively), compared to the earlier emplaced domes viscosities ($< 8 \times 10^5$ Pa s), might be related to either a silica enrichment of the feeding magmas, a lower temperature of the feeding magmas, or a larger amount of crystals in the magmas. Since there is no signature of high-Si contents in spectral data, the two latter hypotheses are favored.

Another possible explanation related to the magma rise speed and source dike dimensions is proposed by [Lena et al. \(2013\)](#). The previous classification scheme put the domes of category B (Arago 2 and 3) in a distinct rheologic group compared to categories H1 and E1. Domes of groups E1 and H1 would display high effusion rates but low edifice

Table 1

Morphometric properties of the eight Arago domes. Given the irregular shape of the domes, both a minimum and maximum height and radius were measured.

Dome	Coordinates		Height (m)		Mean radius (m)		Slope (°)		Area (km ²)		Volume (km ³)	
	Latitude (°)	Longitude (°)	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Arago 1	7.70	22.07	103	159	2550.25	3146.25	2.46	2.98	17.81	24.90	0.70	1.65
Arago 2	7.57	21.55	320	398	9767.25	12021.50	1.96	1.92	240.02	395.85	31.97	60.23
Arago 3	6.16	19.94	245	397	8411.25	12048.00	1.69	1.90	212.90	425.41	18.15	60.35
Arago 4	8.52	21.21	38	114	2748.50	7113.25	0.82	0.93	18.36	129.19	0.30	6.04
Arago 5	8.95	20.94	46	94	3276.50	4805.50	0.81	1.15	31.87	77.38	0.52	2.28
Arago 6	9.27	20.76	84	129	3666.00	4068.25	1.41	1.84	35.45	43.11	1.18	2.24
Arago 7	11.29	24.11	85	105	2427.75	2859.25	2.01	2.12	18.42	22.53	0.52	0.90
Arago 8	4.70	22.72	116	213	3435.50	5240.00	2.54	2.62	26.96	57.97	1.43	6.12

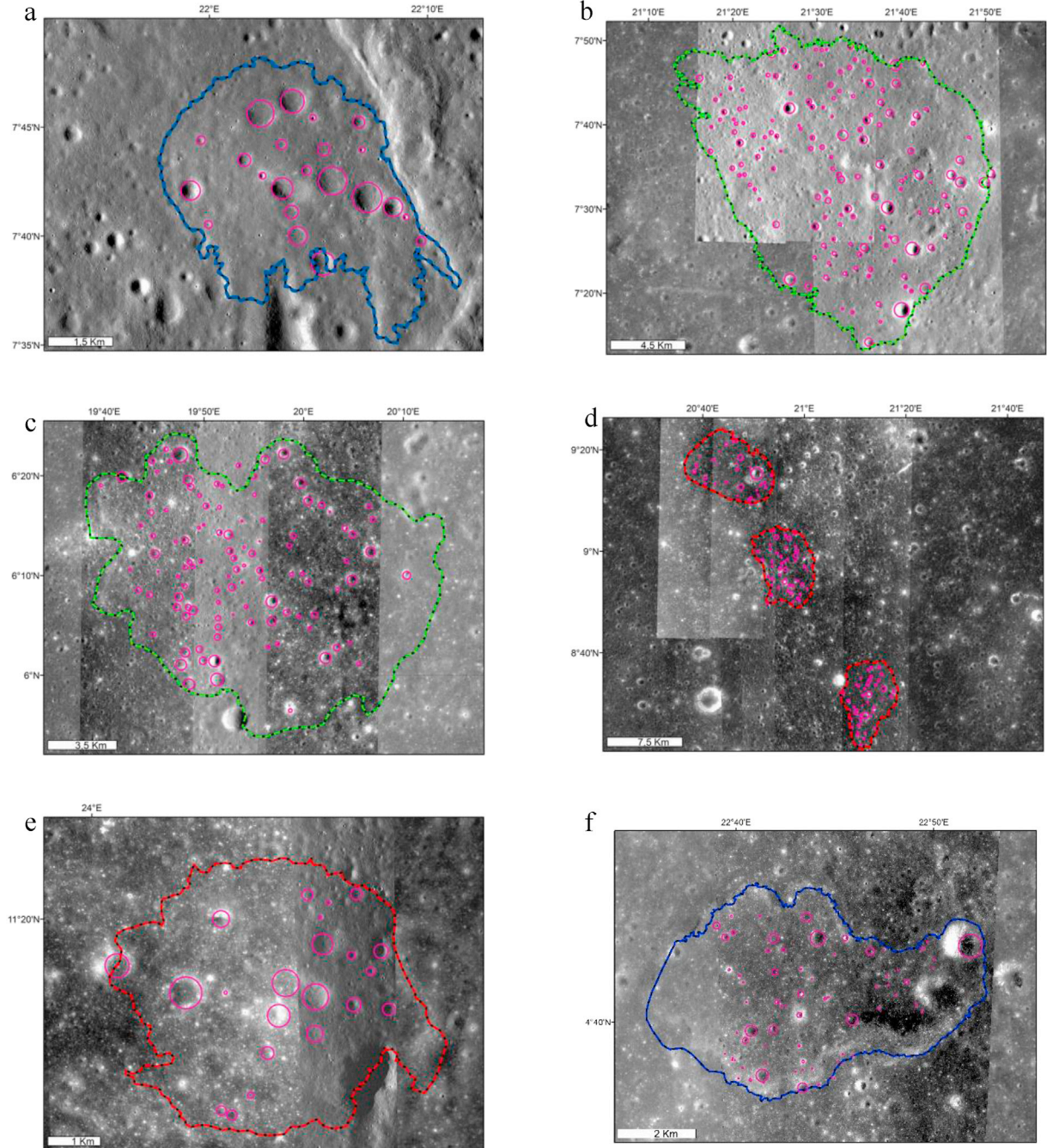


Fig. 6. A to F: Crater counts on the NAC images of domes (A1, A2,A3, A4-6, A7 and A8 respectively). Craters (in magenta) were counted on the entire dome area with the exception of dome A8 as the western portion of the dome is not covered by NAC imagery (counted area is indicated by a yellow outline).

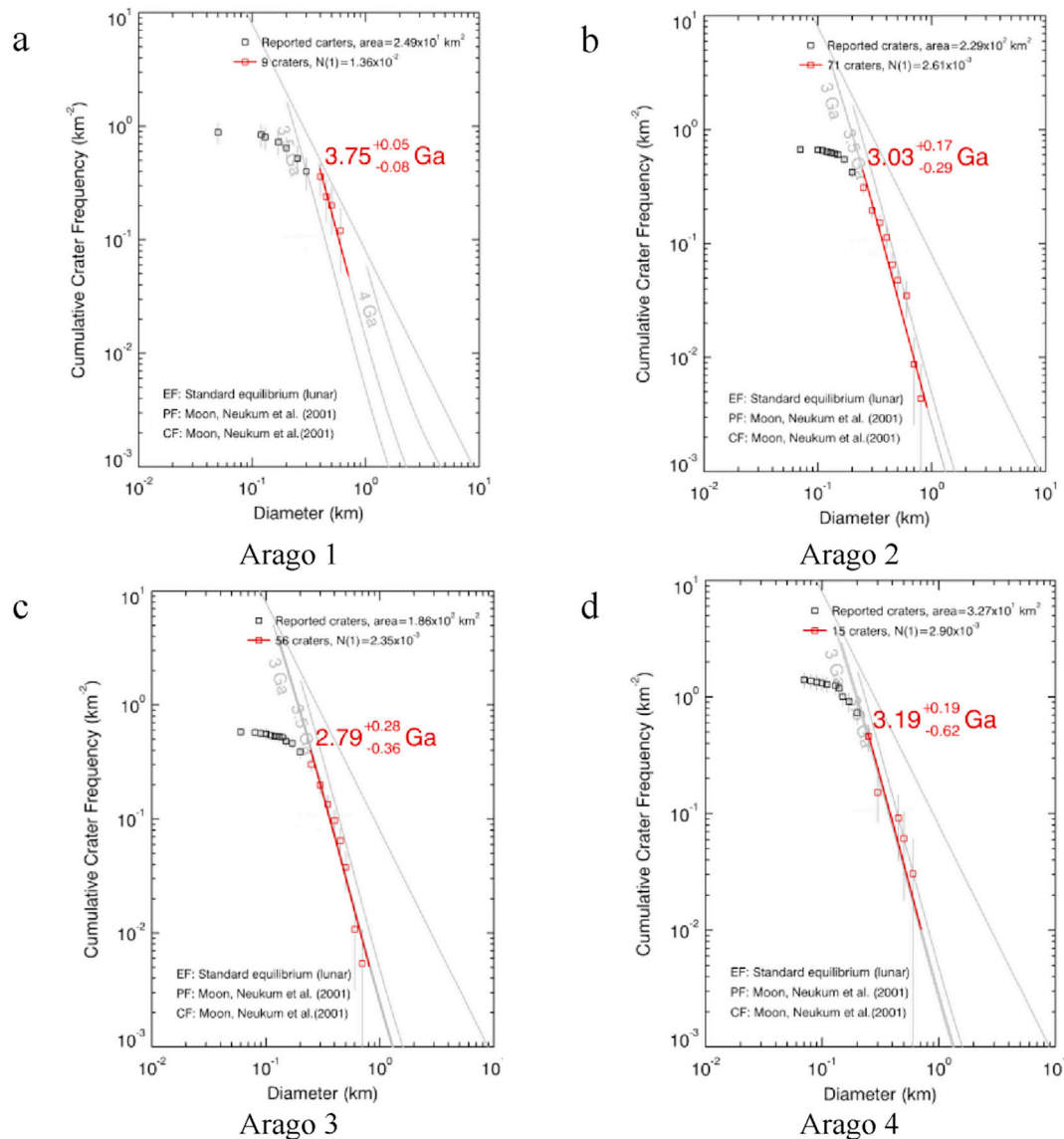


Fig. 7. A to H: Corresponding isochrones of Fig. 6 and fitted ages.

volumes consistent with a short duration of the effusion process. Category B domes would be emplaced by much larger and deeper feeder dikes, and would be characterized by lower lava effusion rates but larger erupted lava products, implying a longer duration of the effusion process (Lena et al., 2013).

The proposed chronology in this work, assuming the domes to be monogenetic, disagrees with previous work (Lena et al., 2013) which suggested that Arago 2 and 3 could belong to class D (i.e., polygenetic domes). We did not find evidence of plural pulses of volcanism for domes Arago 2 and 3, as reported for some of the Marius Hills domes (Whitford-Stark and Head, 1977) or the Gruithuisen and Mairan domes (Wilson and Head, 2003; Glotch et al., 2011). Crater distributions on the Arago 1 to 5 and 7 domes (Fig. 7) are rather uniform and do not show any slope break indicative of major resurfacing events. The lack of alignment over a single isochrone for dome Arago 6 can be explained by the superimposition of IMPs which induced partial resurfacing. Crater distributions for dome Arago 8 also do not align over an isochrone, suggesting a possible partial resurfacing, and possibly hints at a polygenetic origin (Fig. 5c). Zhang et al. (2018) proposed that Arago 8 could correspond to a lava plateau rather than a dome, but the conclusions would remain similar in both cases, in terms of emplacement from a fissural eruption in

more than a single episode.

The general lack of distinguishable units and the relatively small sizes of the other Arago domes are therefore consistent with an emplacement during a single episode. The relatively short duration of lava eruption ($T_e < 7$ years) also make a single stage dome-forming eruptive event plausible.

4.3. Implications for the region history

A previous geologic mapping work within Mare Tranquillitatis was performed by Kodama and Yamaguchi (2003). This study revealed that the Arago domes are located onto two specific geologic units: domes Arago 2 and 3 are located in the Tr3 unit (medium-Ti unit, dated at 3.67 Ga), whereas domes Arago 1, 4, 5, 6, 7 and 8 are located in the Tr4 unit (the youngest regional unit dated at 3.59 Ga with also the higher-Ti contents of the mare) (Fig. 3). The region contains wrinkle ridges, one sinuous rille, and numerous extensional grabens that surround, or are part of, the Tr4 unit (Fig. 3). Almost all the IMPs present in Mare Tranquillitatis are located within the Tr4 unit and they partly overlap some domes, suggesting that they are among the youngest volcanic deposits in the area. Based on these observations, the following chronology is

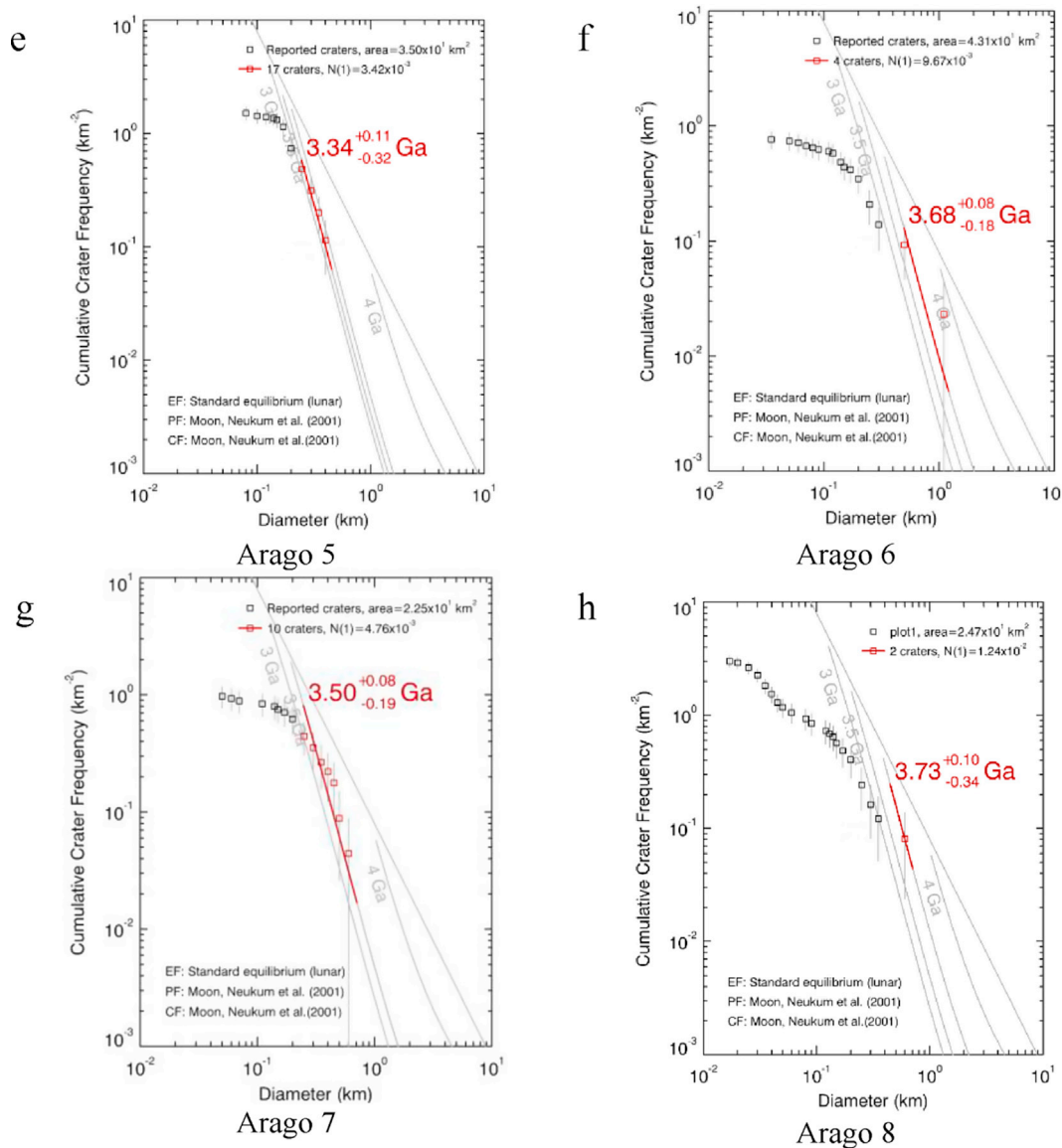


Fig. 7. (continued).

Table 2

Modelled rheologic properties of the eight Arago domes. Mean heights shown in row 4 were used to perform calculations.

Dome	Slope (°)	Diameter (km)	Height (m)	τ (yield strength, Pa)	η (plastic viscosity, Pa.s)	E (lava effusion rate, m ³ /s)	Te (eruption duration, years)
Arago 1	2.72	5.70	131	6.34 × 10 ³	8.01 × 10 ⁵	43.05	0.87
Arago 2	1.94	21.79	359	1.25 × 10 ⁴	4.05 × 10 ⁶	229.84	6.36
Arago 3	1.79	20.46	321	1.06 × 10 ⁴	2.75 × 10 ⁶	226.63	5.49
Arago 4	0.88	9.86	76	1.23 × 10 ³	1.57 × 10 ⁴	222.40	0.45
Arago 5	0.98	8.08	70	1.28 × 10 ³	1.71 × 10 ⁴	162.18	0.27
Arago 6	1.63	7.73	106.50	3.09 × 10 ³	1.42 × 10 ⁵	97.62	0.56
Arago 7	2.07	5.29	95	3.59 × 10 ³	2.05 × 10 ⁵	51.14	0.44
Arago 8	2.58	8.68	164.50	6.57 × 10 ³	8.71 × 10 ⁵	79.52	1.51

proposed.

The emplacement of Mare Tranquillitatis basalts started more than 3.85 Ga ago (during the Early Imbrian) with the emplacement of the Tr1 to Tr3 geologic units (Kodama and Yamaguchi, 2003). The Tr4 unit was emplaced during the Late Imbrian (3.59 Ga ago) and the region was overlain by the small, effusive domes labeled Arago 1 and 8. Unit Tr4 represent the latest, large-scale effusive mare event visible in the Arago region. Domes Arago 4 to 7 are contemporary (3.59 Ga) or younger than Tr4 and might mark the end of the Tr4 mare emplacement with effusive

eruptions along local fractures. Domes Arago 2 and 3, the largest domes with intermediate slopes, were emplaced at the beginning of the Erastosthenian (~2.9 Ga ago), possibly from more evolved, more crystal-rich, or cooler magmas. The late formation of these two domes suggests long-lived volcanism in the region, which is however located outside the PKT, but is still associated with a GRS Thorium anomaly (Lawrence et al., 2003) (Fig. 2d).

Almost all IMPs are located on the Tr4 unit, likely representing a waning stage of dike emplacement in the region (e.g., Wilson and Head,

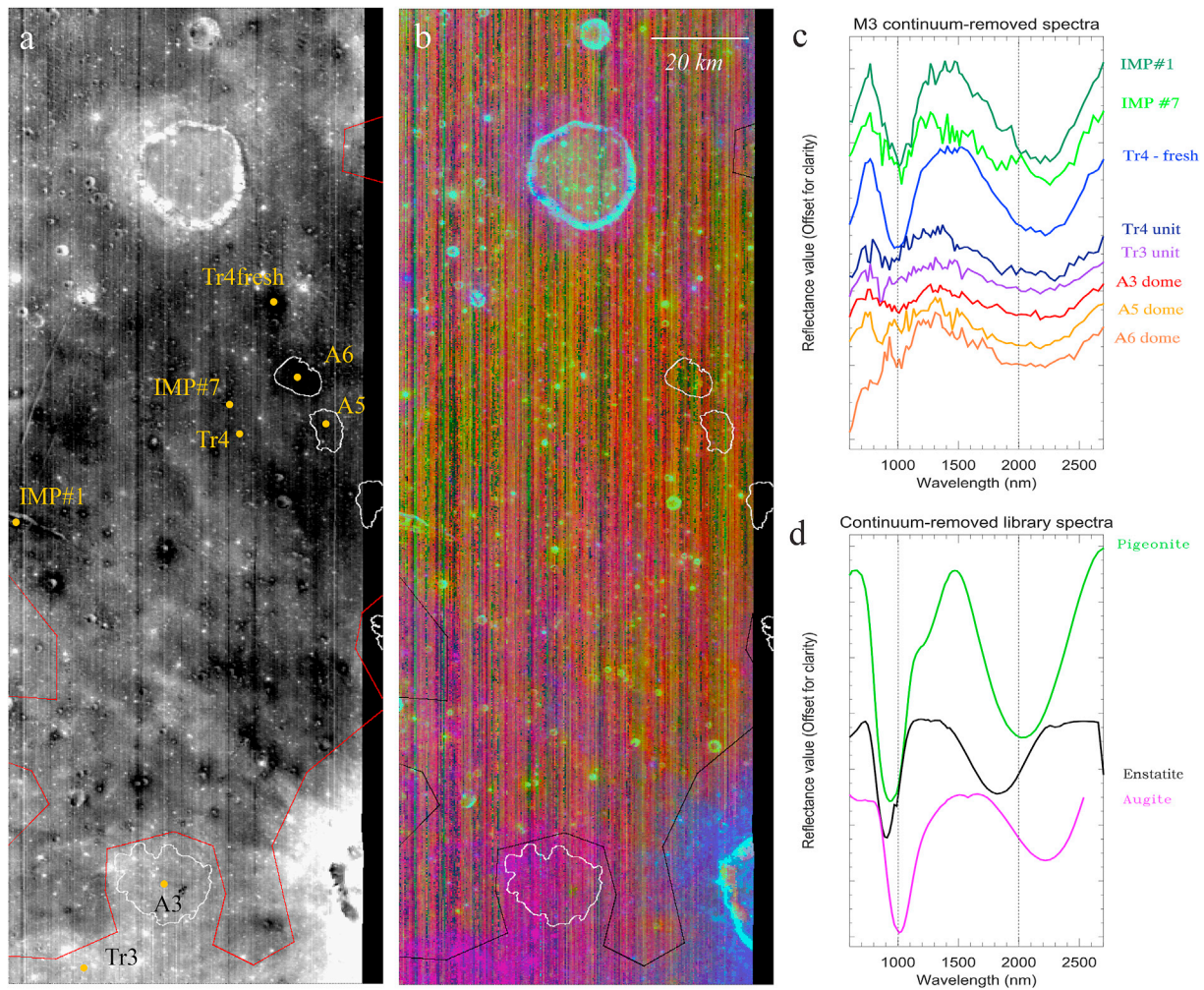


Fig. 8. M³ spectral analysis of the Arago domes and surroundings (M³ stripe M3G20090607T025544 from OP2C1). A. – M³ 2900-nm mosaic (stretched values: 0.119–0.147). Outlines of the domes are shown in white, the limit between mare unit Tr4 and Tr3 (to the south, bottom part of the image) is indicated by a thick red line. The locations of the spectra shown in figure C) are shown by orange dots. B. – M³ RGB composite of spectral parameters from [Martinot et al., \(2018\)](#) (R = 2 μ m band center, stretched values: 1.900–2,400 nm; G = 2 μ m band depth, stretched values: 0.04–0.09; B = reflectance at 1,580 nm, stretched values: 0.07–0.10). C. – M³ continuum-removed spectra of domes A3, A5 and A6, of nearby mare units Tr3, Tr4 (including a spectra taken over a fresh, small crater, labeled Tr4fresh) and of IMPs #1 (Sosigenes) and #7. M³ spectra are 3 $\times$ 3 pixel averages with the exception of the spectrum taken over IMP#7 which is a single pixel spectrum (as the IMP is smaller in size than a 3x3 pixel average). Data were processed using the algorithm developed by [Martinot et al., \(2018\)](#) (see section 2). Reflectance values were offset and vertical bars were placed at 1000 and 2000 nm for clarity. D. – Continuum-removed pyroxene spectra from the RELAB spectral library, processed with the same routine (method of [Martinot et al., 2018](#)), are shown for comparison. RELAB respective samples ID are: AG-TJM-010/C1AG10, DL-CMP-008/BKR1DL008, and DH-MBW-005/BKR1DH005 for augite, pigeonite, and enstatite. Reflectance values were offset and vertical bars were placed at 1000 and 2000 nm for clarity.

2017). IMPs were initially interpreted as young (<1 Ga) volcanic features because of their optical immaturity and the low number of superimposed impact craters (e.g., [Braden et al., 2014](#)). Recent studies suggest that IMPs more likely formed concurrently or slightly later than the surrounding mare lava flows (e.g., [Garry et al., 2012](#); [Wilson and Head, 2017](#); [Zhang et al., 2018](#)). Similarly to the recently discovered Ring-Moat Dome Structures (RMDs) commonly observed in lunar mare regions, IMPs are indeed best explained as the result of basaltic lava flows inflation during their cooling phases (e.g., [Garry et al., 2012](#); [Zhang et al., 2017](#); [Wilson and Head, 2017](#); [Qiao et al., 2018](#); [Wilson et al., 2019](#)). IMPs formation mechanisms are controversial, but magma degassing below a solidified lava crust as observed in Hawaiian lava lakes (e.g., [Wilson and Head, 2017](#)) could induce cracking of the crust, resulting in the extrusion of gas-rich magmatic foams to form hummocky mounds. We favor this mechanism to explain the extrusion of IMPs onto the Tr4 unit, at the time of its emplacement or shortly after. At locations where the gas might have been able to escape from magmatic intrusions, domes could have been formed instead of IMPs. Four IMPs are, however,

situated on the dome Arago 6 (Fig. 5b), implying that they should have formed after the dome emplacement. The small size of the IMPs (tenth of meters) observed in this unit could be due to the important thickness of the Tr4 unit, minimizing the quantity of extruding foam and resulting in small IMPs. We infer the domes Arago 4 to 7 to be contemporaneous or younger than unit Tr4, similarly to the IMPs. Because of the gaps in age between the three groups of domes in the Arago region, it is however possible that the IMPs and dome formation periods overlapped, and that some of the IMPs predate the youngest domes Arago 2 and 3, which might have been fed by a different, deeper magma source (see section 4.2).

The emplacement of the Diamondback sinuous rille, Arago domes and regional IMPs can therefore be interpreted as the result of intrusive dikes reaching different depths within the cooling lunar crust (e.g., [Zhang et al., 2018](#); [Wilson and Head, 2017](#)). Dikes breaching the surface and erupting may create high flux effusive eruptions such as sinuous rilles, whereas dikes just reaching the near surface may extrude small amounts of lava, producing small domes. As the lunar crust cools and thickens over

time, dikes may have remained trapped at shallow depths and have undergone gas exsolution. Their accumulation, and the local foam extrusion, would eventually form the IMPs. Dikes remaining trapped at depth may result in grabens and pit chains instead. This interpretation however implies the existence of a magma storage beneath the mare to support outgassing lava eruption. Although Mare Tranquillitatis is not associated with any lunar mascon, positive Bouguer anomalies observable from the GRAIL dataset are located ~90 km southeast of dome Arago 2, just below dome Arago 8 (the “Lamont” anomaly) and another 40 km south of dome Arago 7 (the northern extension of the Lamont anomaly) (Fig. 2e). This may indicate the presence of large-scale magmatic intrusions at depth (Zuber et al., 2013). The Lamont gravity anomaly was previously interpreted as a buried impact structure associated with both intrusive and extrusive volcanism (e.g., Evans et al., 2016; Zhang et al., 2018). These previous studies show that ancient buried impact structures such as Lamont, may be accompanied by dense dike intrusions, crater-concentric and crater-radial patterns of alignment of volcanic features such as cones, domes, rilles, linear or arc-shaped ridges and grabens (e.g. Zhang et al., 2018). Fig. 3 shows that most wrinkle ridges and extensional grabens in the Arago region are indeed concentric to the Lamont anomaly, whereas domes Arago 1, 2, 4, 5, 6 and 8 are aligned radially to the Lamont structure. As it is the case for several domes, some of the IMPs are aligned radially, and might represent the surface manifestation of large-scale, intrusive dikes likely connected to the Lamont structure.

4.4. The Arago region, a suitable landing site for future science and ISRU missions?

The Arago region exhibits a diversity of volcanic features and deposits, suggesting a long-lived magmatic history, possibly spanning over one billion years. These volcanic features were formed by a range of volcanic eruption styles, including flooding volcanism (rille and mare units), small shield volcanoes (domes) and outgassing process related depressions (vents and IMPs). The long duration of volcanism (>1 Ga) is rather unusual for such locations outside of PKT. As lunar volcanism provides a window into the thermal and compositional evolution of the Moon, such regions contain a diverse and long geological record that is key to future scientific advances (e.g., NRC, 2007; Crawford et al., 2012; Jaumann et al., 2012). The Arago region therefore represents a high scientific interest site for future missions of exploration (e.g., Flahaut et al., 2012). In addition, the Arago region offers an interesting target for future ISRU activities. The Arago region corresponds to a Thorium positive anomaly (Fig. 2d) which might be of interest for nuclear power, although thorium still lacks applications compared to uranium (e.g., Crawford, 2015). But most significantly, the region is dominated by Ti-rich units, with estimates of 13.7 ± 0.7 wt% TiO_2 for the mare unit Tr4 (where most of the domes are located) and 12.6 ± 0.8 wt% TiO_2 for unit Tr3 (where Arago 2 and 3 are located) (Kodama and Yamaguchi, 2003). Unit Tr4 extends further south to the Tranquility Base where Apollo 11 astronauts collected soils with 7.5–12 wt% TiO_2 (e.g., Beaty, 1979; Lucey et al., 2006) which is mainly accommodated in the mineral ilmenite (FeTiO_3). Together with Oceanus Procellarum and the western part of Mare Imbrium, Mare Tranquillitatis is one of the few vast regions of high ilmenite concentrations on the Moon. Ilmenite has long been recognized as a key resource for oxygen production through mineral reduction (e.g., Cutler and Krag, 1985; Gibson and Knudsen, 1985; Taylor and Carrier, 1992; Anand et al., 2012). A number of reductants is possible (e.g., Gibson and Knudsen, 1985), although hydrogen seems to be favored. Ilmenite reduction ranks high among ISRU processes to be tested and demonstrated with upcoming missions, owing to its relative simplicity and technological readiness (e.g., Anand et al., 2012; Crawford, 2015). As noted by Anand et al. (2012), this process also yields a metal product (titanium) which may be of additional economic importance. The deposit of interest for mining appears rather wide and locally homogeneous from orbit, and the predicted content has been confirmed with ground truth.

Given its combined scientific interest and potential for resource

utilization, we argue that the Arago region is a key candidate site for a future exploration mission. It would also offer the unique opportunity to fly over Tranquility Base 50 years after the first human walked on the Moon.

5. Conclusion

Eight volcanic domes in the Arago region, western Mare Tranquillitatis, were reported and characterized in this paper. Despite various morphometric characteristics, most domes are interpreted as single stage eruptive events, effusive in nature. High resolution observations, combined with the results of this study, suggest the following geological history of Mare Tranquillitatis. The emplacement of mare basalts older than 3.85 Ga (regional unit Tr1) was followed by the emplacement of additional lava flows (Tr2 to Tr5), volcanic domes and superimposed IMPs, a period which lasted for more than 1 Ga. Of the eight Arago mare domes, the youngest ones (Arago 2 and 3) display distinct rheological parameters and may originate from a distinct, deeper source. The Arago region is located outside of the PKT, but it is associated with a positive Thorium anomaly, which could explain its exceptional volcanic history and complexity. The Arago region unique geologic record holds clues to the Moon's thermal and magmatic evolution. In addition, the Arago region corresponds to one of the few ilmenite-rich lunar mare deposits presenting a strong potential for future ISRU activities.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pss.2020.104901>.

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